Linear Algebra 2026 Homework 3

Due: October 16, 2026

Problems should be solved in English and calculation steps are required. For all T/F questions, you don't need to justify your answer.

All notations can be found in the book "Linear Algebra — the fourth edition" by Stephen H. Friedberg, Arnold J. Insel, Lawrence E. Spence. If you cannot locate them, feel free to contact the TAs.

Problem 1 [12 pts]. Label the following statements as true or false.

(a) The zero vector space has no basis.

(b) Every vector space that is generated by a finite set has a basis.

(c) Every vector space has a finite basis.

(d) A vector space cannot have more than one basis.

(e) If a vector space has a finite basis, then the number of vectors in every basis is the same.

(f) The dimension of $\mathsf{P}_n(F)$ is n.

(g) The dimension of $\mathsf{M}_{m\times n}(F)$ is $m+n$.

(h) Suppose that V is a finite-dimensional vector space, that S_1 is a linearly independent subset of V, and that S_2 is a subset of V that generates V. Then S_1 cannot contain more vectors than S_2.

(i) If S generates the vector space V, then every vector in V can be written as a linear combination of vectors in S in only one way.

(j) Every subspace of a finite-dimensional space is finite-dimensional.

(k) If V is a vector space having dimension n, then V has exactly one subspace with dimension 0 and exactly one subspace with dimension n.

(l) If V is a vector space having dimension n, and if S is a subset of V with n vectors, then S is linearly independent if and only if S spans V.

Problem 2 [10 pts]. Let W denote the subspace of R^5 consisting of all the vectors having coordinates that sum to zero. The vectors

$$u_1 = (2,-3,4,-5,2), \qquad u_2 = (-6,9,-12,15,-6),$$
$$u_3 = (3,-2,7,-9,1), \qquad u_4 = (2,-8,2,-2,6),$$
$$u_5 = (-1,1,2,1,-3), \qquad u_6 = (0,-3,-18,9,12),$$
$$u_7 = (1,0,-2,3,-2), \qquad u_8 = (2,-1,1,-9,7)$$

generate W. Find a subset of the set $\{u_1, u_2, \ldots, u_8\}$ that is a basis for W.

Problem 3 [8 pts]. Let u and v be distinct vectors of a vector space V. Show that if $\{u, v\}$ is a basis for V and a and b are nonzero scalars, then both $\{u+v, au\}$ and $\{au, bv\}$ are also bases for V.

Problem 4 [10 pts]. Let $v_1, v_2, \ldots, v_k, v$ be vectors in a vector space V, and define $\mathsf{W}_1 = \text{span}(\{v_1, v_2, \ldots, v_k\})$, and $\mathsf{W}_2 = \text{span}(\{v_1, v_2, \ldots, v_k, v\})$.

(a) Find necessary and sufficient conditions on v such that $\dim(\mathsf{W}_1) = \dim(\mathsf{W}_2)$.

(b) State and prove a relationship involving $\dim(\mathsf{W}_1)$ and $\dim(\mathsf{W}_2)$ in the case that $\dim(\mathsf{W}_1) \neq \dim(\mathsf{W}_2)$.

Problem 5 require knowledge of the sum and direct sum of subspaces. You can find the definitions in the last homework.

Problem 5 [20 pts].

(a) Prove that if W_1 and W_2 are finite-dimensional subspaces of a vector space V, then the subspace $\mathsf{W}_1 + \mathsf{W}_2$ is finite-dimensional, and $\dim(\mathsf{W}_1 + \mathsf{W}_2) = \dim(\mathsf{W}_1) + \dim(\mathsf{W}_2) - \dim(\mathsf{W}_1 \cap \mathsf{W}_2)$. (10 pts)

Hint: Start with a basis $\{u_1, u_2, \ldots, u_k\}$ for $\mathsf{W}_1 \cap \mathsf{W}_2$ and extend this set to a basis $\{u_1, u_2, \ldots, u_k, v_1, v_2, \ldots, v_m\}$ for W_1 and to a basis $\{u_1, u_2, \ldots, u_k, w_1, w_2, \ldots, w_p\}$ for W_2.

(b) Let W_1 and W_2 be finite-dimensional subspaces of a vector space V, and let $\mathsf{V} = \mathsf{W}_1 + \mathsf{W}_2$. Deduce that V is the direct sum of W_1 and W_2 if and only if $\dim(\mathsf{V}) = \dim(\mathsf{W}_1) + \dim(\mathsf{W}_2)$. (10 pts)

Problem 6 [8 pts]. Label the following statements as true or false. In each part, V and W are finite-dimensional vector spaces (over F), and T is a function from V to W.

(a) If T is linear, then T preserves sums and scalar products.

(b) If $\mathsf{T}(x + y) = \mathsf{T}(x) + \mathsf{T}(y)$, then T is linear.

(c) T is one-to-one if and only if the only vector x such that $\mathsf{T}(x) = 0$ is $x = 0$.

(d) If T is linear, then $\mathsf{T}(0_\mathsf{V}) = 0_\mathsf{W}$.

(e) If T is linear, then $\text{nullity}(\mathsf{T}) + \text{rank}(\mathsf{T}) = \dim(\mathsf{W})$.

(f) If T is linear, then T carries linearly independent subsets of V onto linearly independent subsets of W.

(g) If $\mathsf{T}, \mathsf{U} : \mathsf{V} \to \mathsf{W}$ are both linear and agree on a basis for V, then $\mathsf{T} = \mathsf{U}$.

(h) Given $x_1, x_2 \in \mathsf{V}$ and $y_1, y_2 \in \mathsf{W}$, there exists a linear transformation $\mathsf{T} : \mathsf{V} \to \mathsf{W}$ such that $\mathsf{T}(x_1) = y_1$ and $\mathsf{T}(x_2) = y_2$.

For Problems 7-9, prove that T is a linear transformation, and find bases for both $\mathsf{N}(\mathsf{T})$ and $\mathsf{R}(\mathsf{T})$. Then compute the nullity and rank of T, and verify the dimension theorem. Finally, use the appropriate theorems in this section to determine whether T is one-to-one or onto.

Problem 7 [5 pts]. $\mathsf{T} : \mathsf{R}^3 \to \mathsf{R}^2$ defined by $\mathsf{T}(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$.

Problem 8 [5 pts]. $\mathsf{T} : \mathsf{P}_2(R) \to \mathsf{P}_3(R)$ defined by $\mathsf{T}(f(x)) = xf(x) + f'(x)$.

Problem 9 [5 pts]. $\mathsf{T} : \mathsf{M}_{n\times n}(F) \to F$ defined by $\mathsf{T}(A) = \mathrm{tr}(A)$. Recall that

$$\mathrm{tr}(A) = \sum_{i=1}^{n} A_{ii}.$$

Problem 10 [10 pts]. In this exercise, $\mathsf{T} : \mathsf{R}^2 \to \mathsf{R}^2$ is a function. For each of the following parts, state why T is not linear.

(a) $\mathsf{T}(a_1, a_2) = (1, a_2)$

(b) $\mathsf{T}(a_1, a_2) = (a_1, a_1^2)$

Problem 11 [10 pts]. Prove that there exists a linear transformation $\mathsf{T} : \mathsf{R}^2 \to \mathsf{R}^3$ such that $\mathsf{T}(1, 1) = (1, 0, 2)$ and $\mathsf{T}(2, 3) = (1, -1, 4)$. What is $\mathsf{T}(8, 11)$?

Problem 12 [10 pts]. Let $\mathsf{T} : \mathsf{P}(R) \to \mathsf{P}(R)$ be defined by $\mathsf{T}(f(x)) = f'(x)$. Recall that T is linear. Prove that T is onto, but not one-to-one.

Problem 13 [10 pts]. Let V and W be finite-dimensional vector spaces and $\mathsf{T} : \mathsf{V} \to \mathsf{W}$ be linear.

(a) Prove that if $\dim(\mathsf{V}) < \dim(\mathsf{W})$, then T cannot be onto.

(b) Prove that if $\dim(\mathsf{V}) > \dim(\mathsf{W})$, then T cannot be one-to-one.

The following definitions are used in Problems 14, 15.

Definition 1. Let V be a vector space, and let $\mathsf{T} : \mathsf{V} \to \mathsf{V}$ be linear. A subspace W of V is said to be **T-invariant** if $\mathsf{T}(x) \in \mathsf{W}$ for every $x \in \mathsf{W}$, that is, $\mathsf{T}(\mathsf{W}) \subseteq \mathsf{W}$. If W is T-invariant, we define the **restriction of T on W** to be the function $\mathsf{T}_\mathsf{W} : \mathsf{W} \to \mathsf{W}$ defined by $\mathsf{T}_\mathsf{W}(x) = \mathsf{T}(x)$ for all $x \in \mathsf{W}$.

Problems 14, 15 assume that W is a subspace of a vector space V and that $\mathsf{T} : \mathsf{V} \to \mathsf{V}$ is linear.

Problem 14 [8 pts]. Prove that the subspaces $\{0\}$, V, $\mathsf{R}(\mathsf{T})$, and $\mathsf{N}(\mathsf{T})$ are all T-invariant.

Problem 15 [8 pts]. If W is T-invariant, prove that T_W is linear.

Problem 16 [12 pts]. A function $\mathsf{T} : \mathsf{V} \to \mathsf{W}$ between vector spaces V and W is called **additive** if $\mathsf{T}(x + y) = \mathsf{T}(x) + \mathsf{T}(y)$ for all $x, y \in \mathsf{V}$. Prove that if V and W are vector spaces over the field of rational numbers, then any additive function from V into W is a linear transformation.

The following exercise requires familiarity with the definition of *quotient space* given in Problem 11 of homework 2.

Problem 17 [10 pts]. Let V be a vector space and W be a subspace of V. Define the mapping $\eta : \mathsf{V} \to \mathsf{V}/\mathsf{W}$ by $\eta(v) = v + \mathsf{W}$ for $v \in \mathsf{V}$.

(a) Prove that η is a linear transformation from V onto V/W and that $\mathsf{N}(\eta) = \mathsf{W}$.

(b) Suppose that V is finite-dimensional. Use (a) and the dimension theorem to derive a formula relating $\dim(\mathsf{V})$, $\dim(\mathsf{W})$, and $\dim(\mathsf{V}/\mathsf{W})$.

Problem 18 [16 pts]. Let $\alpha_1, \alpha_2, \cdots, \alpha_m$ ($m \geq 2$) be vectors in a vector space over $\mathbb{R}$. α_m is not a zero vector. Prove that the following statements are equivalent:

1. for any real numbers $k_1, k_2, \cdots, k_{m-1}$, the vectors

$$\beta_1 = \alpha_1 + k_1\alpha_m,\ \beta_2 = \alpha_2 + k_2\alpha_m,\ \cdots,\ \beta_{m-1} = \alpha_{m-1} + k_{m-1}\alpha_m$$

   are linearly independent.

2. The vectors $\alpha_1, \alpha_2, \cdots, \alpha_m$ are linearly independent.